MCQ 1: Understand and Choose

Five teaching questions with one answer. Choose one answer per question; then explain one This chapter format does not reproduce the official format of the EAM.

1. Rate of x^2 between 1 and 2:

   A. 2
   B. 3
   C. 1
   D. 4

2. If $f(1) = 4$, $f\prime(1) = 2$, tangent:

   A. $y = 2x + 2$
   B. $y = 4x + 2$
   C. $y = 2x + 4$
   D. $y = x + 3$

3. The derived number is

   A. a function
   B. an area
   C. an abscissa
   D. a real if the derivative exists

4. For x^2 in 3, derived number:

   A. 9
   B. 3
   C. 6
   D. 12

5. The approximation of $f(a + h)$ is

   A. $f(a)f\prime(a)$
   B. $f(a) + hf\prime(a)$
   C. $f(a) + f\prime(h)$
   D. $hf(a)$

MCQ 2: understand and choose

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1. $\sqrt{x}$ in 0 is

   A. Derivable derivative 0
   B. Derivable derivatives 1
   C. Non-derivable
   D. Undefined

2. $\dfrac{|h|}{h}$ for $h < 0$ is

   A. -1
   B. 1

C. 0

D. h

3. A tangent of slope 0 is

A. Vertical

B. oblique

C. Unable

D. Horizontal

4. Position in m and time in s: derivative unit

A. m^2

B. $\mathrm{m\,s}^{-1}$

C. $\mathrm{s\,m}^{-1}$

D. m

5. For derivative a, replace $f(a+h)$ by $f(a)+hf\prime(a)$ is generally

A. an exact identity

B. a definition of h

C. Local approximation

D. Always false even for an affine

True or false?

1. A tangent can never cross a curve.

2. The absolute value function is derivatable in 0.

3. An affine function $mx+p$ a for derivative number m in all real.

Personal review

I give the domain and h not zero; I read a slope; I use the pitch point; I distinguish exact and Approached.
For each objective, indicate: acquired without help; succeeded with help; to rework. Write an example and a mistake to avoid.